# Mobile App Intervention to Promote Breast Cancer Screening in Underserved Immigrant Women

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## Overview

### Background

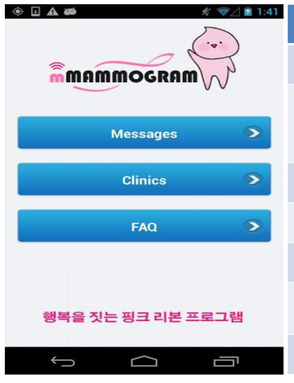
- Korean American women have one of the highest breast cancer mortality rates, and one of the lowest breast cancer screening rates. There is little research and intervention development focused on Korean American women and breast cancer.

### Research Objective

- To assess the effectiveness of the *m*Mammogram intervention, designed to promote knowledge of breast cancer screening and receipt of mammograms

mMAMMOGRAM

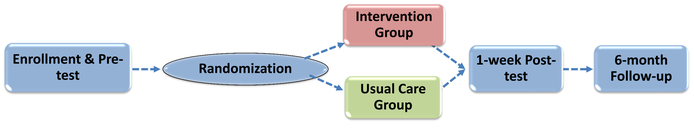
- 7 day mobile phone app-based program ; 8 to 21 text, visual, and video messages per day; Pink Ribbon incentive; Health navigator service




| Day | Theme |
|---|---|
| 1 | Introduction to breast cancer |
| 2 | Introduction to breast cancer screening methods (SBE, CBE, and Mammogram) |
| 3 | "Knowledge day" |
| 4 | "Health accessibility day" |
| 5 | "Culture day" |
| 6 | Clinic information and summary |
| 7 | Review |

### Methods

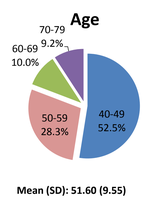
- *Research design*: Two-arm randomized controlled trial and community based participatory research (CBPR)

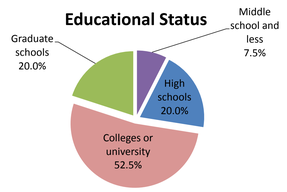


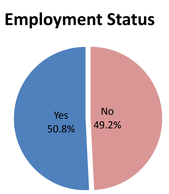

- *Participants*: 120 Korean American immigrant women, aged 40-79, living in Minnesota, who did not get a mammogram within the past two years
- Data analysis: independent-samples *t*-test, chi-square test, and mixed-design *ANOVA* (within-subject and between-subject factors)

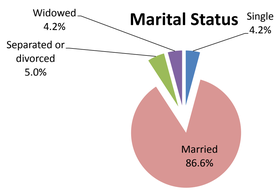
## Findings

### Sociodemographic Characteristics (n = 120)

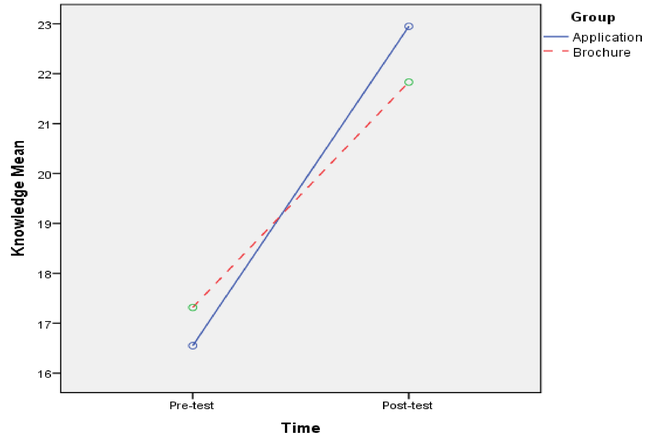


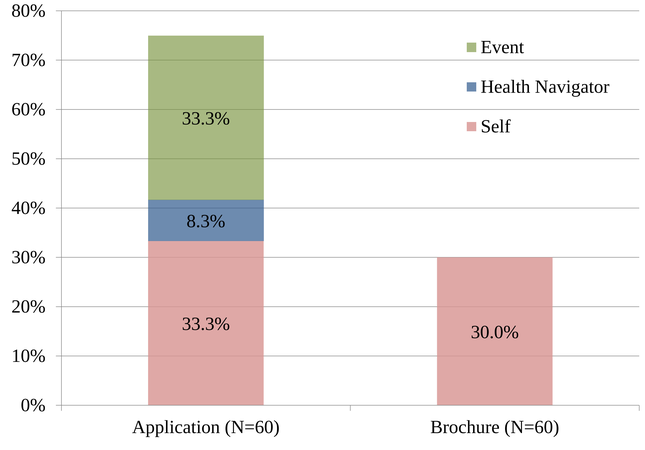




### Changes in Screening Knowledge after Intervention (n = 120)




### Receipt of Mammogram after Intervention (n = 120)




## Conclusion & Implications

### Conclusion

- *Knowledge*: No significant between-group difference, but the intervention group reported increased knowledge after the intervention.
- *Receipt of Mammogram*: Significant between-group differences in receipt of mammogram; more intervention group participants received a mammogram post-intervention compared to the control group.

### Implications

- The combination of mobile application and health navigation services could synergistically complement each other to encourage receipt of a mammogram.
- A mobile phone-based health intervention could be an effective method in minority populations with high breast cancer burdens, using tailored messages that cover broad content areas and overcome restrictions to place and time of delivery.

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